

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/358715420>

Prevalence of sport injuries in Olympic combat sports. A cross-sectional study examining one Olympic period

Article in *The Journal of sports medicine and physical fitness* · February 2022

DOI: 10.23736/S0022-4707.22.13334-7

CITATIONS

10

8 authors, including:



Christophe Lambert

Witten/Herdecke University

27 PUBLICATIONS 258 CITATIONS

[SEE PROFILE](#)



Daniel Lachmann

17 PUBLICATIONS 112 CITATIONS

[SEE PROFILE](#)

READS

887



Ramona Ritzmann

University of Freiburg

136 PUBLICATIONS 1,904 CITATIONS

[SEE PROFILE](#)



Nikolaos Malliaropoulos

Barts Health NHS Trust

145 PUBLICATIONS 3,243 CITATIONS

[SEE PROFILE](#)



Modified defense reaction reduces biomechanical and myoelectrical ACL injury risk factors in elite Judo

Christophe Lambert, Janine Riesterer, Marlene Mauch, Maxime Lambert, Jochen Paul & Ramona Ritzmann

To cite this article: Christophe Lambert, Janine Riesterer, Marlene Mauch, Maxime Lambert, Jochen Paul & Ramona Ritzmann (2022): Modified defense reaction reduces biomechanical and myoelectrical ACL injury risk factors in elite Judo, Journal of Sports Sciences, DOI: [10.1080/02640414.2022.2080160](https://doi.org/10.1080/02640414.2022.2080160)

To link to this article: <https://doi.org/10.1080/02640414.2022.2080160>



Published online: 26 May 2022.



Submit your article to this journal [↗](#)



View related articles [↗](#)



View Crossmark data [↗](#)

Modified defense reaction reduces biomechanical and myoelectrical ACL injury risk factors in elite Judo

Christophe Lambert^{a,b}, Janine Riesterer^a, Marlene Mauch^a, Maxime Lambert^b, Jochen Paul^{a##} and Ramona Ritzmann^{a##}

^aDepartment of Functional Diagnostics, Rennbahnklinik, Muttentz, Switzerland; ^bDepartment of Trauma and Orthopedic Surgery, University of Witten/Herdecke, Cologne Merheim Medical Centre, Cologne, Germany

ABSTRACT

In judo, an anterior cruciate ligament (ACL) injury is the most severe injury an athlete could experience. Most ACL ruptures occur when defending against an *osoto-gari* attack. This study aims to identify ACL risk factors during *osoto-gari* defence and implement a *modified osoto-gari defence reaction*, which is assumed to improve myoelectric patterns and ameliorate critical biomechanical risk factors for ACL injuries. Twenty-six elite judokas were enrolled in the cross-over trial (female: 6; male: 20). 3D kinematics and force dynamometrics were combined with electromyographical recordings to assess the effects of the *common* and the *modified osoto-gari defence reaction*. Compared to the *common osoto-gari defence reaction* (maximal knee flexion: $29 \pm 12^\circ$; maximal valgus: $10 \pm 5^\circ$; maximal valgus moment: 58 ± 17 Nm; peak internal rotation: $9 \pm 5^\circ$), the *modified osoto-gari defence reaction* showed significantly reduced knee angles ($31 \pm 10^\circ$ $p < 0.05$; $1 \pm 0^\circ$ $p < 0.05$; 31 ± 9 Nm $p < 0.05$; $3 \pm 0^\circ$ $p < 0.05$). The myoelectric activity of the hamstring increased ($+5\pm\%$ to $+27\pm\%$, $p < 0.05$) in the modified compared to common defence reaction. The *modified osoto-gari defence reaction* reduced critical biomechanical risk factors and increased hamstring myoelectric activity. We recommend the implementation of the *modified osoto-gari defence reaction* in judo practice and seek to evaluate its long-term effectiveness in decreasing ACL injury incidences in elite judo.

ARTICLE HISTORY

Accepted 4 May 2022

KEYWORDS

ACL; neuromuscular; reflex; kinematics; sports medicine; elite sports; athlete

Introduction

Judo is a widespread combat sports, with athletes practicing judo in over 150 countries (Murata, 2005). At the 2012 London Olympics, judo was the sport with the second-largest country distribution, only behind athletics (Engebretsen et al., 2013). Epidemiological studies reveal that the knee joint is one of the most affected articulations, with a quote of 30% of all injuries (Frey et al., 2019; Kim et al., 2015). Independent of the weight category, an anterior cruciate ligament (ACL) rupture is the most severe injury with high clinical relevance, causing up to one year of down time and a persistently reduced performance in 68% of affected athletes (Majewski et al., 2006). Akoto et al. provides numerical data on 4,659 judoka indicating that approximately 10% suffer an ACL injury during their career (Akoto et al., 2017) in combat while in direct or indirect contact situations with opponents (Koshida et al., 2010). Koshida et al. identified the *osoto-gari* as the technique with the highest incidence of ACL injuries (Koshida et al., 2010). The *osoto-gari* attack is a widely applied throwing technique. The throw is achieved by a major outer leg reap which exposes the athlete in defence to a high risk of experiencing articular damage (Koshida et al., 2010).

With regard to the prevention of ACL injuries, it is paramount to determine the osteokinematics of primary injury mechanisms and to establish alternative judo techniques coupled with preventive

training strategies. However, insufficient evidence exists to identify which factors account for the injury mechanisms and how injury avoidance can be achieved. Previous studies identify a specific combination of critical osteo-kinematic trajectories as the primary risk factors for ACL injuries: i) an increased valgus stress (Koga et al., 2010), ii) a knee flexion of approximately 30° (Li et al., 1999; Utturkar et al., 2013) simultaneous with an iii) internal rotation which increase tibiofemoral contact forces (Navacchia et al., 2019). These trajectories are accompanied by excessive joint torques which diminish the mechanical congruence of the lateral and medial tibiofemoral joints and cause an elongation or tear of the ACL (Koga et al., 2010; Li et al., 1999; Navacchia et al., 2019; Utturkar et al., 2013). Using electromyography, muscle weakness reflected by a delayed and diminished neuromuscular innervation has also been identified as being paramount within the hazard assessment of ACL injuries (Renstrom et al., 1986; Serpell et al., 2015). As specified in isolated monoarticular and complex motor tasks, reduced medial and lateral hamstring synergies (Serpell et al., 2015) lead to an imbalance in quadriceps-hamstring activation (Ruas et al., 2019). This can facilitate anterior-tibial translation and thus gradually increases ACL stress throughout ligamentous elongation (Bencke et al., 2018). However, it is unclear as to what extent these mechanisms are presented in judo-specific techniques and if the injury-relevant parameters can be noticeably modified by technical modification. The latter aspect of injury avoidance involves retraining simple, alternative movement

patterns in the defence reaction which may reduce the aforementioned risk factors i) – iii). Clarity regarding this research question could make a decisive contribution to the implementation of a new judo-specific injury prevention programme aiming to minimize the number of severe injuries (Koshida et al., 2010).

The aim of this study was to examine the biomechanical and neuromuscular mechanisms underlying in injury that occurs while defending against a *common osoto-gari* attack (Koshida et al., 2010) compared to a *modified osoto-gari defence reaction*. For that purpose, we analysed relevant osteokinematic and neuromuscular factors using the gold standard methodology of electromyography and three-dimensional (3D) kinematics. It is hypothesized that athletes applying the *common osoto-gari defence reaction* display higher knee joint loadings accompanied with reduced muscle excitation levels in the musculature encompassing and stabilising the knee joint compared to that of the *modified osoto-gari defence reaction*. It is postulated that the *modified osoto-gari defence reaction* can therefore reduce critical biomechanical risk factors.

Material and methods

Study design

We used a repeated measures cross-over design. Participants performed the *common* and the *modified osoto-gari defence reaction* in a random and counterbalanced order during full sparring. Random allocation sequence was achieved by sequentially numbered containers. Both techniques are graphically and technically described in Figure 1; for movement description and technical validity see (Koshida et al., 2010). Note that the *modified osoto-gari defence reaction* was developed consensually in cooperation with the team doctors and trainers of the national teams and the judoka of the Olympic teams. The *modified osoto-gari defence reaction* was created with the focus on a stable stance, reduced thorax back lean and reduced contact time and force between the opponents to allow for greater knee joint control.

Subjects gave written informed consent to the experimental procedure following the latest revision of the Declaration of Helsinki that was approved by the local Ethics Committee.



Figure 1. Schematic of the *osoto-gari* attack and defence reaction (white) for the common *osoto-gari* defence reaction (left) compared to the modified defence reaction (right).

Table 1. Weight classes as the indicator of morphotype.

Weight class	Female (n)	Male (n)
Heavyweight	1	4
Half heavyweight	1	3
Middleweight	1	4
Half middleweight	0	3
Lightweight	3	3
Half lightweight	0	3
Extra lightweight	0	0

Inclusion criteria consisted of an age of >18 years, fit for competition without any acute injuries, no previous ACL injury on the attacked knee and no diseases or medical restrictions. The exclusion criteria were an age below 18 years, injuries, diseases, or any medical limitation.

Study population

Twenty-six elite judokas (6 female and 20 male athletes) were enrolled in the study; all were competing in the International Judo Federation World Tour and the Olympic qualification (mean age: 23.5 ± 3.5 years; height: 176 ± 8.3 cm; weight: 78.5 ± 17.1 kg). The studied population won a total of 12 medals at World Championships and 27 medals at European Championships. Weight classes are described in Table 1.

The sample size calculation was based on the effect sizes observed in a pilot experiment with $n = 5$. The *a-priori* power analysis (G*Power V 3.1.9.2) revealed a sample size of $n = 25$ for an effect size of $d = 0.3$, a power ($1 - \beta$) of 0.85 and an $\alpha < 0.05$ for the parameter angular excursions, torque and integrated EMG. The case number calculation refers to the calculation of an analysis of variance (ANOVA) with repeated measurements.

To check for reproducibility, test-retest reliability estimates were computed by the intra-class correlation coefficient (ICC) using a one-way random single measure model for the pilot sample of $n = 5$ with n items for the repetitions within the conditions “common” and “modified”. Outcomes given by Cronbachs α according were satisfied at levels of 0.937–0.989.

Protocol

A 10-min warm-up (running, tappings, bouncing and pairwise selected judo warm-up techniques) was followed by movement familiarisation to consolidate a reliable and efficient common and modified *osoto-gari* defence reaction. The test trials consisted of a minimum of six valid trials for each defence reaction as illustrated in Figure 1: i) the *common osoto-gari defence reaction* (Kujala et al., 1995) and ii) the *modified osoto-gari defence reaction*.

From initial contact, knee joint loading is associated with disadvantageous changes, increasing the risk of a knee joint injury. The *osoto-gari* attack is executed by the attacking judoka with a sweeping leg movement that makes direct contact with the defending judoka's leg. In the common *osoto-gari* defence reaction, the defending judoka's limbs are nearly parallel and the back is bent backwards. This results in a knee valgus and internal rotation stress on the attacked knee. For movement description and technical validity see, (Koshida et al., 2010).

For the modified *osoto-gari* defence reaction, the defending judoka pulls the unattacked limb backwards when the initial contact of the attacking judoka's limb is felt on the knee, while simultaneously bending the upper body forward. This creates a more stable position for the defending judoka's knee and prevents knee valgus and internal rotation stress.

The athletes performed the trials while only wearing tight and short pants with a naked torso to make the technique and 3D markers visible (Ishii et al., 2018). Each athlete performed the throwing techniques with a weight-category-matched opponent of similar height and weight. They performed the *osoto-gari* according to their usual sparring drills and with a magnitude of force they would apply in competition. Validity was given if the *osoto-gari* throwing technique was performed as described in the study by Imamura et al., (2006) and if there was efficiency in defending the attacking judoka. Consistency and reproducibility of the technical requirements were visually controlled during the familiarisation phase prior to data collection.

Assessment methods

3D kinematics, kinetics and inverse dynamics computation applied to the lower and upper limb were combined with electromyographical recordings to assess movement trajectories, forces and the myoelectric activity of the musculature stabilising the knee joint for both the *common* and *modified defence reaction* against the *osoto-gari* attack (Koshida et al., 2010).

Biomechanical recordings

Horizontal and vertical ground reaction forces occurring in both reactions were measured using a three-dimensional force plate (Kistler, Winterthur, CH) applying a measuring frequency of 1000 Hz. The foot of the support leg of the tested judoka (Figure 1) was positioned on the force plate when performing the movement (Pucsok et al., 2001). Subjects were instructed to focus on the technique by looking straight ahead to ensure a normal posture and defence pattern. A trial was considered valid if the force plate was fully loaded and solely by the support leg of the measured judoka. This could be verified by analysing the vertical component of the ground reaction forces.

Kinematic analysis was performed at a sampling frequency of 200 Hz using a ten-camera 3D-motion analysis system (Vantage, Vicon Motion Systems Ltd., Oxford, UK) according to (Hood et al., 2020). In order to assess joint kinematics, a conventional gait model was applied (Baker et al., 2017). As required by the plug-in gait model, the retro-reflective markers were placed on the following anatomical landmarks (left and right): anterior superior iliac spine, posterior superior iliac spine, thigh, medial and lateral knee joint centre, shank, tibia, lateral and medial malleolus, heel, fifth metatarsal bone and hallux. Those markers were used to generate a 3D-model of the lower body, from which three joint angles were calculated (ankle, knee and hip). Marker trajectories as well as ground reaction forces were low-pass filtered at 15 Hz (Butterworth, fourth order) prior to calculating joint angles and joint moments. All

calculations were performed in an integrated software package (Bodybuilder 3.6, Vicon Motion Systems Ltd., Oxford, UK) and Python. Three-dimensional joint rotations were determined using Euler angles with a flexion/extension, adduction/abduction and internal/external rotation sequence. External joint moments were calculated in the distal segment coordinate system with a standard inverse dynamic approach by integrating kinetic data from the force plate sampled at 1000 Hz using the Golem model (BR, 2013). The joint angles and moments were calculated for each measured time frame, resulting in curves of joint angles and moments against time. The curves were averaged over repetitions of each condition for each subject, respectively.

Myoelectrical recordings

Myoelectrical recordings were executed with surface electromyography (EMG). Wireless electrodes (Myon, Schwarzenberg, Switzerland) were placed over the rectus femoris (RF), vastus lateralis (VL), vastus medialis (VM), biceps femoris (BF) and semitendinosus (ST) muscles, according to the European recommendations for EMG (SENIAM – Surface Electromyography for the non-invasive-assessment of muscles) (Hermens et al., 2000). The longitudinal axes of the electrodes were in line with the presumed direction of the underlying muscle fibres. The inter-electrode resistance was reduced to $<2.5 \Omega$ by shaving and lightly abrading the skin. Signals were sampled with 2000 Hz and band-pass filtered (20 Hz to 450 Hz). Subjects performed isometric maximal voluntary contractions (MVC) for each recorded muscle. The highest EMG values were used for data normalisation per subject (EMG maximum peak ± 25 ms). The MVCs were executed according to (Roelants et al., 2006) and (Wiley & Damiano, 1998), while being performed isometrically against resistance and held for 3 s with 1-min rest in between. Body position during MVCs was strictly controlled and standardised by supervision of the authors.

Data analysis

Trials were considered to be valid if i) each reflexive marker was visible by a minimum of three out of 10 cameras ii) and only the foot of the measured judoka was located on the force plate. The initial leg contacts of the judo opponents served as a reference (Figure 2) to time point 0 for the data analysis. The initial contact was assessed visually with two cameras for each participant and each trial. The criterion was that the opponents touched their legs, as this is considered the beginning of relevant time for the knee injury (Ruas et al., 2019). Kinematic, dynamometric and electromyographic data were analysed before the event as a reference ($-100-0$ ms) and after ($0-250$ ms) the event. The tibiofemoral movement in the sagittal, frontal and transversal trajectories, as well as the maximal joint excursions, the time interval to the maximum joint excursion (Scattone-Silva GCL et al., 2012), peak velocities (Kaminski & Gentile, 1986), and joint torques were assessed using inverse dynamics (Hood et al., 2020) as key variables associated with ACL ruptures (Li et al., 1999; Navacchia et al., 2019; Renstrom et al., 1986; Serpell et al., 2015).

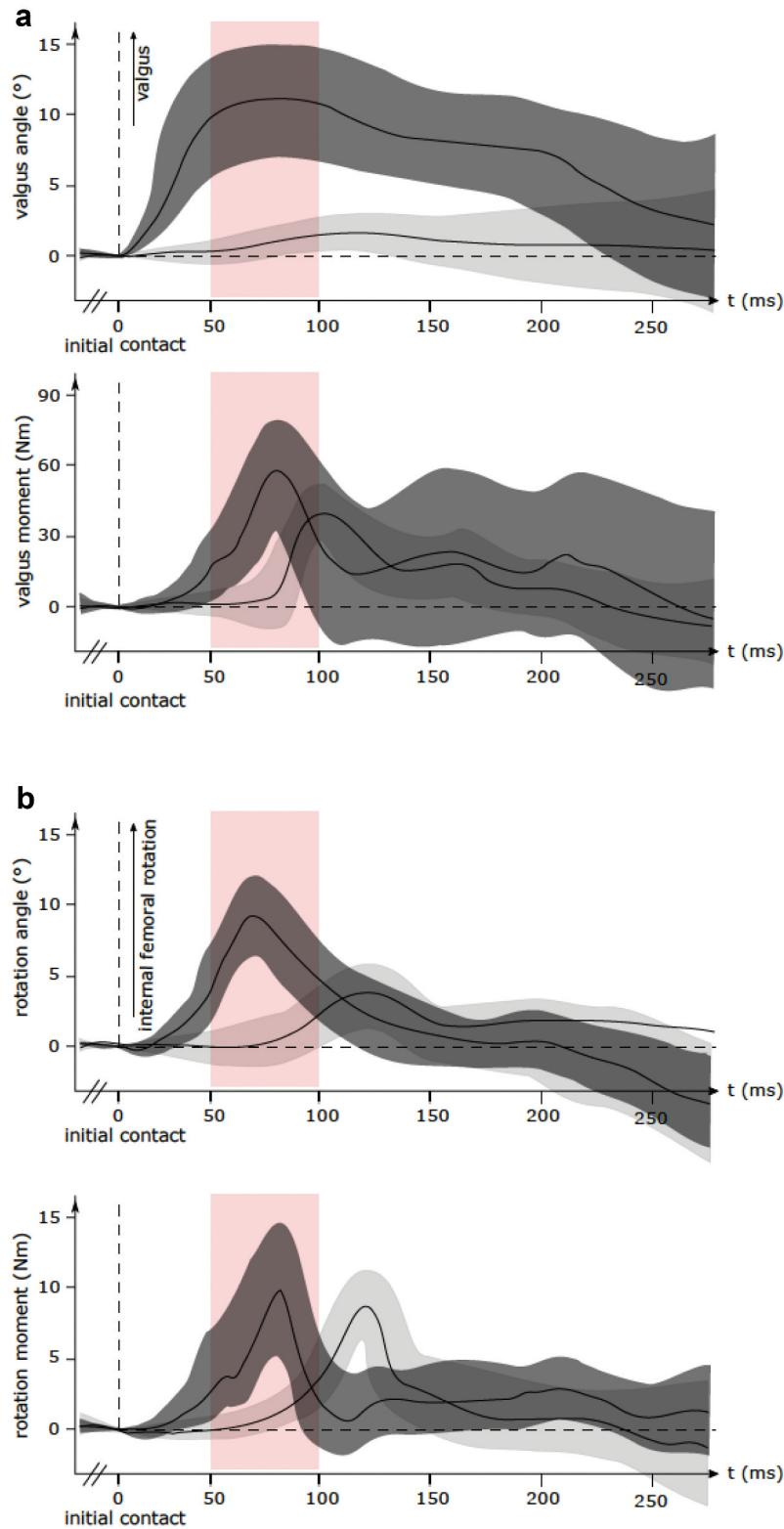


Figure 2. Temporal profiles of frontal and transversal layer trajectories. Knee valgus (A) and rotation (B) angles (femoral internal rotation and tibial internal or external rotation) and estimated joint moments. The *common osoto-gari defence reaction* refers to the graph in dark grey and the *modified osoto-gari defence reaction* is highlighted in light grey. Initial contact indicates the first knee contact of the judoka and the start of the *osoto-gari* attack. 0° indicated the individual knee joint neutral positioning. Values display means and standard deviations with red highlights for the relevant time frame.

The myoelectrical activity was rectified, averaged, integrated (iEMG) and time was normalised for the following phases: fast responses (30–90 ms post-initial contact) (Jennings & Seedhom, 1994; Tsuda et al., 2001) and involving immediate reflex activity

and slow neuromuscular responses (91–250 ms post-initial contact) (Dyhr-Poulsen & Krogsgaard, 2000; Madhavan & Shields, 2011). Afterwards, the iEMG was normalised to the iEMG recorded during MVC of the corresponding muscle. The muscle

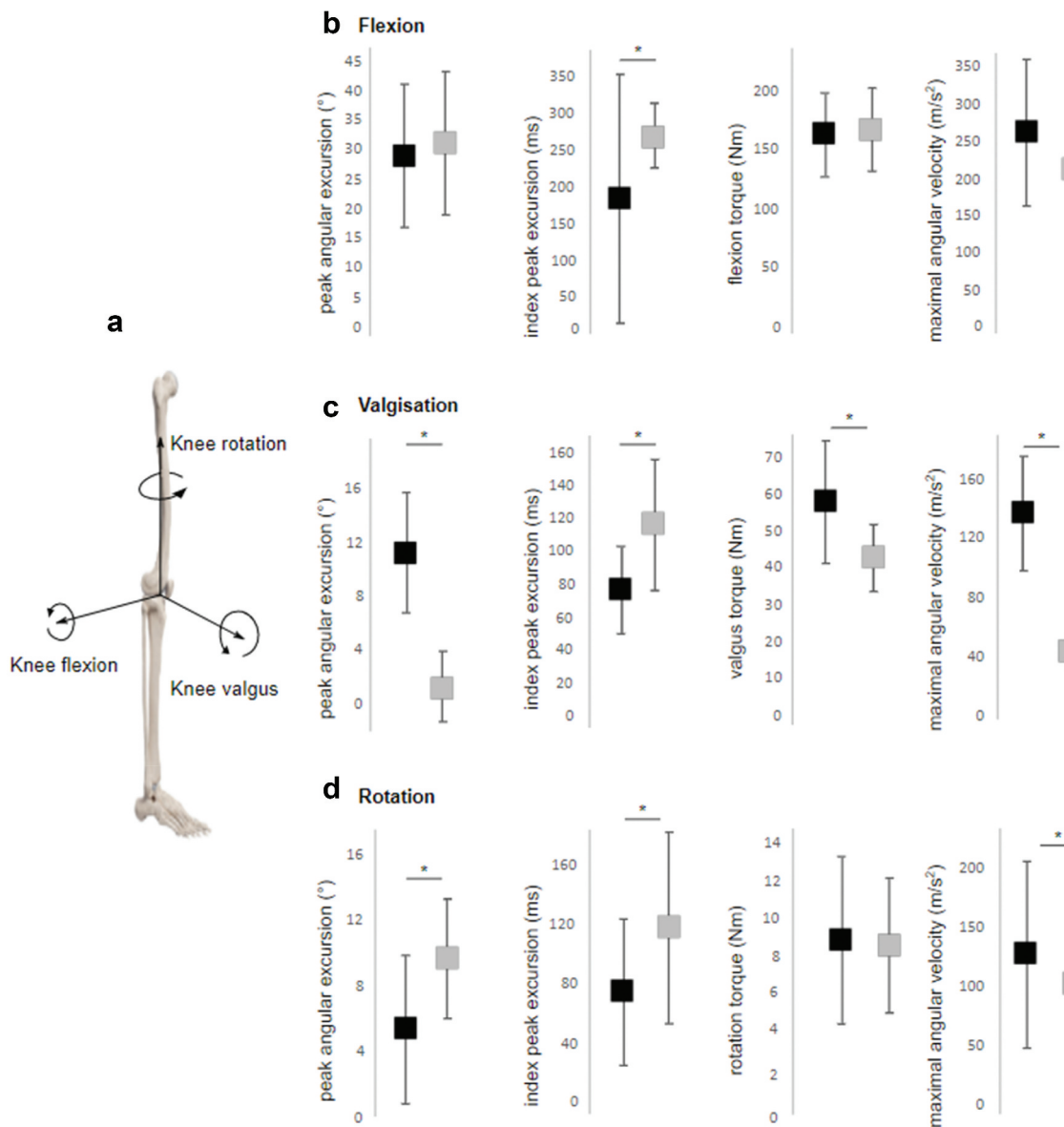


Figure 3. (A) Knee joint motion differentiated by the sagittal, frontal and transversal planes and associated means and standard deviations across athletes for (from left to right) the angular excursions, the index of the angular excursions, the joint torque and the angular velocity of (B) knee flexion, (C) valgization and (D) rotation movements. Symbols in dark grey refer to the *common osoto-gari defence reaction*; the *modified osoto-gari defence reaction* is highlighted in light grey. * indicates a significant difference ($p < 0.05$).

activation hamstring:quadriceps ratio (MA H:Q ratio) was determined according to (Ruas et al., 2019). Quotients were calculated using MVC normalised myoelectric activity of the BF and RF, BF and VM, SMT and RF, SMT and VM for the interval of 30–250 ms after the incidence of initial leg contact.

The data was averaged by the trials and subjects. Outcome assessors were blinded and not aware of the experiment.

Statistics

A repeated measures analyses of variance (ANOVA), with defence reaction [common vs. modified] was used to test for differences. The normality of the data was evaluated with the Kolmogorov-Smirnov test; the data followed a normal distribution. If the assumption of sphericity established by Mauchly's test was violated, the Greenhouse-Geisser correction was used.

The level of significance was set to $p < 0.05$. The false discovery rate was controlled according to the Benjamini-Hochberg-Yekutieli method, a less conservative but still stringent statistical approach conceptualising the rate of type I errors (Y BYH, 1995; Benjamini & Yekutieli, 2005). Partial Eta square ($\eta^2 p$) was used as an estimate of the effect size for the rmANOVA ($\eta^2 p \leq 0.01$ small, $0.01 \geq \eta^2 p \leq 0.14$ medium, $\eta^2 p \geq 0.14$ large effect size).

Statistical tests were executed with SPSS 27.0 (SPSS, Inc., Chicago, IL, USA). Group data are presented as mean value $\pm$ standard deviation.

Results

Data are given for the Judoka being attacked. Eleven athletes had a general malalignment reflected by a varus, two had a valgus and 13 athletes had a neutral position. The

Table 2. Kinematic and kinetic differences between the standard and modified *osoto-gari* defence reaction for the sagittal plane (top), the frontal plane (middle) and the horizontal plane (bottom). Dependent variables are displayed as mean $\pm$ SD values and given as percent changes. The statistics contain rmANOVA results and estimated effect sizes as partial eta squared values (η^2_p).

Parameters	Osoto-gari defence reaction			Statistics	
Knee flexion – sagittal plane	Common (M$\pm$ SD)	Modified (M$\pm$ SD)	Difference (%)	rmANOVA	Effect size η^2_p
Maximal knee flexion ($^\circ$)	28.8 $\pm$ 12.1	31.0 $\pm$ 9.9	+7 $\pm$ 4	$F_{1,19} = 0.992$, $P = 0.33$	0.043
Index of maximal knee flexion (ms)	171 $\pm$ 180	247 $\pm$ 160	+45 $\pm$ 32*	$F_{1,19} = 11.733$, $P = 0.002$	0.348
Maximal angular flexion velocity ($^\circ$ /s)	261 $\pm$ 99	210 $\pm$ 98	-20 $\pm$ 43	$F_{1,19} = 1.195$, $P = 0.286$	0.052
Maximal knee flexion moment (Nm)	162 $\pm$ 35	166 $\pm$ 35	+3 $\pm$ 7	$F_{1,19} = 6.990$, $P = 0.015$	0.241
Knee valgus – frontal plane	Common (M$\pm$ SD)	Modified (M$\pm$ SD)	Difference (%)	rmANOVA	Effect size η^2_p
Maximal knee adduction ($^\circ$)	10.7 $\pm$ 4.5	0.9 $\pm$ 0.2	-91 $\pm$ 20*	$F_{1,19} = 96.097$, $P < 0.001$	0.814
Index maximal knee adduction (ms)	76 $\pm$ 27	116 $\pm$ 40	+52 $\pm$ 31*	$F_{1,19} = 20.691$, $P < 0.001$	0.485
Maximal angular velocity ($^\circ$ /s)	65 $\pm$ 14	42 $\pm$ 10	-65 $\pm$ 37*	$F_{1,19} = 45.073$, $P < 0.001$	0.672
Maximal adduction moment (Nm)	58 $\pm$ 17	31 $\pm$ 9	-53 $\pm$ 26*	$F_{1,19} = 156.510$, $P < 0.001$	0.877
Knee rotation – transversal plane	Common (M$\pm$ SD)	Modified (M$\pm$ SD)	Difference (%)	rmANOVA	Effect size η^2_p
Peak internal rotation ($^\circ$)	9.1 $\pm$ 4.5	3.4 $\pm$ 0.	-63 $\pm$ 42*	$F_{1,19} = 22.06$, $P < 0.001$	0.501
Index peak internal rotation (ms)	74 $\pm$ 49	118 $\pm$ 65	+58 $\pm$ 51*	$F_{1,19} = 27.888$, $P < 0.001$	0.559
Maximal angular velocity ($^\circ$ /s)	127 $\pm$ 79	101 $\pm$ 69	-20 $\pm$ 21*	$F_{1,19} = 7.927$, $P = 0.010$	0.265
Maximal internal rotation moment (Nm)	9 $\pm$ 5	10 $\pm$ 8	+2 $\pm$ 19	$F_{1,19} = 0.26$, $P = 0.872$	0.001

* indicates a significant ($p < 0.05$) main effect of the defence reaction.

Table 3. Differences in myoelectrical activity of six muscles encompassing the knee joint between the standard and modified *osoto-gari* defence reactions. Dependent variables are displayed as mean $\pm$ SD values and given as percent changes. The statistics contain rmANOVA results and estimated effect sizes as partial eta squared values (η^2_p).

Parameters	Osoto-gari defence reaction			Statistics	
EMG (%MVC)	Common (M $\pm$ SD)	Modified (M $\pm$ SD)	Difference (%)	rmANOVA	Effect size η^2_p
<i>M. biceps femoris</i>					
Fast responses (30–90 ms post)	31 $\pm$ 7	44 $\pm$ 8	+13 $\pm$ 6*	$F_{1,19} = 21.071$, $P = 0.002$	0.774
Slow responses (91–250 ms post)	40 $\pm$ 11	61 $\pm$ 9	+19 $\pm$ 12*	$F_{1,19} = 4.339$, $P = 0.03$	0.491
<i>M. semitendinosus</i>					
Fast responses (30–90 ms post)	15 $\pm$ 6	20 $\pm$ 7	+5 $\pm$ 7	$F_{1,19} = 1.729$, $P = 0.14$	0.100
Slow responses (91–250 ms post)	39 $\pm$ 13	67 $\pm$ 14	+27 $\pm$ 19*	$F_{1,19} = 84.619$, $P < 0.001$	0.917
<i>M. rectus femoris</i>					
Fast responses (30–90 ms post)	53 $\pm$ 8	59 $\pm$ 10	+6 $\pm$ 11	$F_{1,19} = 0.793$, $P = 0.41$	0.089
Slow responses (91–250 ms post)	50 $\pm$ 12	71 $\pm$ 12	+18 $\pm$ 19*	$F_{1,19} = 18.681$, $P = 0.008$	0.29
<i>M. vastus lateralis</i>					
Fast responses (30–90 ms post)	11 $\pm$ 9	25 $\pm$ 11	+10 $\pm$ 16*	$F_{1,19} = 7.770$, $P = 0.01$	0.691
Slow responses (91–250 ms post)	41 $\pm$ 13	59 $\pm$ 16	+19 $\pm$ 18*	$F_{1,19} = 50.318$, $P < 0.001$	0.841
<i>M. vastus medialis</i>					
Fast responses (30–90 ms post)	28 $\pm$ 10	39 $\pm$ 13	+13 $\pm$ 15*	$F_{1,19} = 36.229$, $P = 0.009$	0.827
Slow responses (91–250 ms post)	31 $\pm$ 12	54 $\pm$ 14	-23 $\pm$ 14*	$F_{1,19} = 43.099$, $P < 0.01$	0.696

* indicates a significant ($p < 0.05$) main effect of the defence reaction.

tibiofemoral movement during defence in the sagittal, frontal and transversal trajectories are displayed in Figure 2, Figure 3 and Table 2. Changes in myoelectric activity are displayed in Table 3 and Figure 4.

Biomechanics in the common *osoto-gari* defence reaction

The *common osoto-gari defence reaction* led to progressive knee flexion angles concomitant with a valgus and tibiofemoral rotation in the defending Judoka. Peak excursions were typically reached 80 ms to 180 ms after initial contact (Table 2). Clinically critical trajectories with high injury risk profiles (Koga et al., 2010; Li et al., 1999; Navacchia et al., 2019; Utturkar et al., 2013) were demonstrated through a simultaneous repositioning in the transversal and frontal plane in response to leg contact between the defending and attacking judoka: valgus loading increased as referenced by an increase in knee adduction angle ($11 \pm 5^\circ$), peak velocity ($136 \pm 39^\circ$ /s) and peak moment (58 ± 17 Nm)

which was coupled with a tibiofemoral rotation adopted because of an increased externally rotated tibia in relation to the femur ($9.1 \pm 4.5^\circ$) in the presence of high angular peak velocities ($127 \pm 79^\circ$ /s). The knee flexion peaked at $29 \pm 12^\circ$ at a peak velocity of $261 \pm 99^\circ$ /s and a peak moment of 162 ± 35 Nm.

Biomechanics in the modified *osoto-gari* defence reaction

The *modified osoto-gari defence reaction* led to significant changes of movement trajectories (Figure 2, 3) accompanied by an enhancement in neuronal excitation of the musculature encompassing the knee joint (Figure 4): the *modified* compared to the *common defence reaction* significantly reduced the tibiofemoral valgus loading (knee adduction angle $10 \pm 5^\circ$, peak adduction velocity $94 \pm 38^\circ$ /s and peak moment 27 ± 19 Nm) and rotation (knee rotation angle $6 \pm 5^\circ$, peak rotational velocity $25 \pm 21^\circ$ /s).

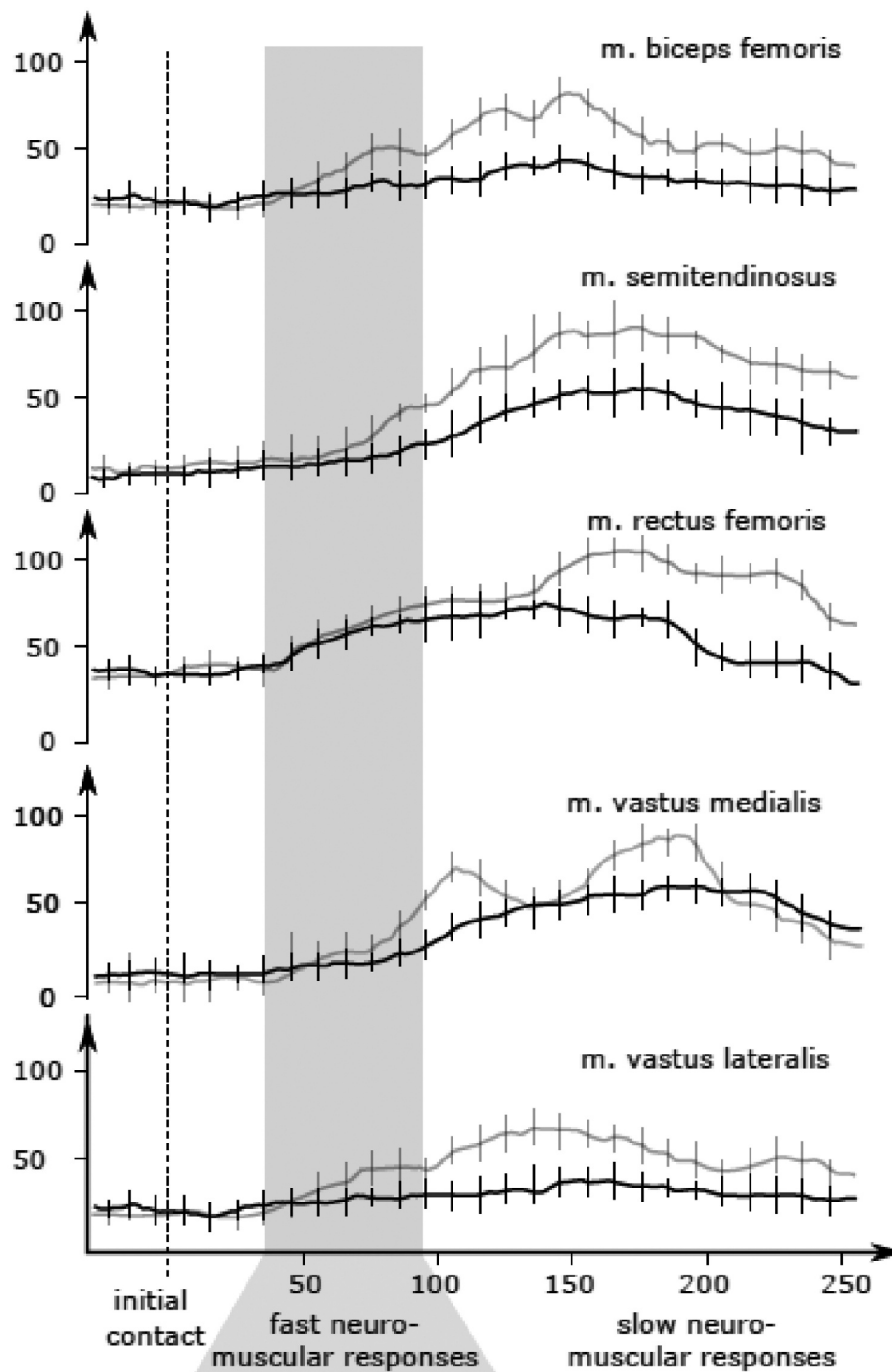


Figure 4. Linear envelopes of myograms from seven recorded muscles. The myograms are averages from three trials across 20 subjects and expressed as relative to the maximum myoelectrical activity. Vertical bars indicate the standard deviations. Curves in dark grey refers to the *common osoto-gari defence reaction* and curves in light grey refer to the *modified osoto-gari defence reaction*. Fast neuromuscular responses entail the interval 30–90 ms after initial contact; slow neuromuscular responses entail the interval 90–250 ms after initial contact.

Neuromuscular parameters in the common osoto-gari defence reaction

The MVC-normalised myoelectric activity of the five recorded muscles is illustrated in Figure 4 and displayed in Table 3. All recorded muscles showed a small level of activation prior to initial contact. The hamstrings showed a moderate increase in activation profiles of fast (30–90 ms) (BF $31 \pm 7\%$ SMT $15 \pm 6\%$ of MVC) and

slow (91–250 ms) neuromuscular responses (BF $44 \pm 8\%$ SMT $61 \pm 9\%$ of MVC) after initial contact. The corresponding peak amplitudes were reached 145 ms (BF) and 175 ms (SMT) after the initial contact. Myoelectrical activity in the knee extensors RF and VM increased progressively until the activation peak was reached 142 ms (RF) and 186 ms (VM) after contact, respectively. The VM remained unaffected and EMG activity did not rise after as

compared to before the common *osoto-gari* attack was initiated. MA H:Q ratios were below 1 and 0.7 ± 0.2 for BF:RF, 0.9 ± 0.2 for BF:VM 0.4 ± 0.1 for SMT:RF, and 0.7 ± 0.2 for SMT:VM.

Neuromuscular parameters in the modified *osoto-gari* defence reaction

All recorded muscles showed a small level of activation prior to initial contact. There were no differences between defence reactions. From initial opponent contact on, knee extensor and flexor muscles immediately had an increase in myoelectrical activity as revealed by significant increases in EMG activity for all recorded in the slow responses and for BF, VL and VM in the fast responses (Table 3). As indicated by the linear envelopes in Figure 4, peak activity was reached 147 ms (BF), 171 ms (SMT), 177 ms (RF), 200 ms (VM) and 142 ms (VL) after contact.

MA H:Q ratios for the *modified osoto-gari* defence reaction were significantly higher compared to the standard reaction. They reached values beyond a ratio of 1 for BF:VM (1.1 ± 0.3 , $p < 0.05$) and SMT:VM (1.0 ± 0.4 , $p < 0.05$) and below 1 for BF:RF (0.8 ± 0.3 , $p < 0.05$) and SMT:RF (0.7 ± 0.3 , $p < 0.05$).

Discussion

This study provides major insights into the injury mechanism of ACL ruptures in judo during full sparring. The *common osoto-gari* defence reaction to the *osoto-gari* attack is associated with elevated risk factors for suffering an ACL injury (Koga et al., 2010; Li et al., 1999; Navacchia et al., 2019; Utturkar et al., 2013). Reduced and delayed EMG activities of knee flexors and extensors, decreased myoelectric H/Q ratios concomitant with a rapid and high magnitude of knee valgus and internal rotation stress leads to combat circumstances with a high ACL injury hazard for the athlete in defence (Koga et al., 2010; Koshida et al., 2010; Li et al., 1999; Navacchia et al., 2019; Utturkar et al., 2013). The *modified osoto-gari* defence reaction reduces the risk factors for suffering an ACL injury, demonstrated by immediately elevated myoelectric activities, a shift in the myoelectric H/Q ratio towards 1 and knee joint trajectories of minor injury risk concomitant with diminished angular torques. Strategic avoidance of an ACL injury can be therefore achieved by familiarisation of alternative defence techniques with a diminished risk hazard.

Kinematics in the ACL injury mechanism

From an osteokinematics perspective, current literature identifies various biomechanical variables such as valgus angle, valgus moment, internal knee rotation and flexion angles of the knee joint as important risk factors for ACL injuries. One of the most discussed risk situation has been described as a valgus of 10° to 15° (Fukuda et al., 2003), which occurs simultaneously with an anterior rotation (10°) of the tibia (Navacchia et al., 2019) during a small knee flexion (Podraza & White, 2010). The *common defence reaction* to an *osoto-gari* attack demonstrates this disadvantageous combination of high-risk kinematic trajectories

which occurs as valgus stress and rotational force applied to the participant's leg by the opponent's action prior to the throwing phase causes damage to the knee joint and elongation stress to the ACL. The current data underline the results of (Koshida et al., 2010), which show that being attacked by an opponent with *osoto-gari* is a primary cause of ACL injuries in collegiate Judo. Subjects who were injured by being attacked with *osoto-gari* self-reported that they had felt the valgus collapse in the knee joint at the time of incident as well as a forceful rotation transmitted by the opponent (Koshida et al., 2010). High joint torques as demonstrated by our study and displayed in Table 2 are related to elongation stress for the ACL and the risk of tear. The consistency between sport disciplines demonstrates the aforementioned injury pattern of knee valgus and rotation is often associated with muscle weakness and reduced activation profiles (Shin et al., 2011), which are demonstrated as follows.

Neuromuscular parameters in the ACL injury mechanism

The forces of the muscles encompassing the knee joint reach up to 5,000 N (Grapar Zargi et al., 2017) when fully activated and, therefore, their synergistic interplay, activation pattern and intensities are paramount for securing ligamentous (Paschos et al., 2010) or cartilage structures of the knee joint during severe injuries. The neuronal control of knee extensors and flexors via spinal and supraspinal (Tsuda et al., 2001) circuitries which allow for immediate (30–90 ms) and long-latency (<90 ms) responses (Dyhre-Poulsen & Krogsgaard, 2000; Madhavan & Shields, 2011) are imperative for adequate joint torques in injury situations (Beard et al., 1994). For the standard defence reaction, myoelectric activity normalised to MVC remained submaximal between 10% and 50%, showing a slow and moderate rise in activation profiles after initial contact. The activation profile of the knee flexors did not exceed 50% of its maximum amplitude during MVC, which is comparably small with regard to the risky circumstance athletes are exposed to (Koshida et al., 2010). The latter aspect is particularly surprising as the hamstrings are known to counteract the anterior translation of tibia plateaus in situations where the ACL is mechanically stressed and elongated throughout external loading (Renstrom et al., 1986; Serpell et al., 2015). We therefore postulate that the muscles encompassing the knee joint may not be activated sufficiently to stabilise and secure the knee joint structure during the *common defence reaction* of an *osoto-gari* attack. In addition, antagonistic muscle coordination, as reflected by MA H/Q ratios remaining below 1 (Ruas et al., 2019), indicates a smaller knee flexor activation as compared to that of knee extensors (Renstrom et al., 1986). Such factor indicates that this muscle imbalance is disadvantageous for knee stability as it favours laxity and anterior tibial shifts. It is noteworthy to mention that the preparatory muscle activity prior to opponent contact was similar for both defence reaction. This indicates that differences in myoelectrical activation after initial contact were due to the technical modifications only.

The modified osoto-gari defence reaction for knee joint injury avoidance

The *modified osoto-gari defence reaction* significantly reduced valgus and rotation stress on the knee as compared to the *common osoto-gari defence reaction*. Angular excursions, velocities and joint torques were diminished. It is noteworthy to mention that previous studies did not observe a significant increase in ACL injuries throughout isolated application of valgus stress only (Fleming et al., 2001). Markolf et al. states that the highest forces acting on the knee are caused by simultaneous anterior tibial torque and a valgus moment with the knee almost extended (Markolf et al., 1995). Joint kinematics and dynamics during the *modified osoto-gari defence reaction* were found in trajectories commonly postulated to be without an elevated risk to experience an ACL injury (Simonsen et al., 2000). Concomitantly, myoelectric activity increased immediately and to a greater magnitude in the musculature, providing greater knee stability (Tsuda et al., 2001). MA H/Q ratios furthermore underline the shift in neuromuscular activation profiles towards a protective hamstring strategy as indicated by ratios > 1 (Ruas et al., 2019). Hamstring synergism induced by BF and SMT activation was previously shown to be efficient in protection from ACL tears (Serpell et al., 2015). It maintains knee stability, providing synergistic action to the ACL by preventing excessive anterior displacement and internal rotation of the tibia. It is important to mention that in both antagonistic muscle groups (knee extensors and flexors) encompassing the knee joint, an increase in myoelectrical activation is documented in the *modified defence reaction* as compared to the *common reaction*. Previous studies have provided scientific evidence that co-activation of the medial and lateral hamstrings and quadriceps muscles reduces ACL stress, demonstrated by reduced ACL elongation (Serpell et al., 2015) through articular stiffening and augmented joint rigidity (Hirokawa et al., 1991).

Despite the multicausality of ACL injuries, the aforementioned factors predisposing individuals to a greater injury risk of the ACL can be minimised by adequately familiarising elite judoka with the *modified osoto-gari defence reaction*.

Limitations

The study has limitations which should be considered. First, experimental evidence is provided for one judo technique only. Although the *osoto-gari* defence reaction is reported as the manoeuvre with the highest ACL risk, further judo techniques do exist that also bear risks for ACL injuries. Future studies should therefore focus on additional techniques and their defence reactions. Second, because this study was designed to develop a new defence strategy in judo, prospective long-term trials, including injury epidemiology during training and competitive fights, are needed to present to what extent this *modified osoto-gari* defence reaction reduces the incidence of ACL injuries compared to the common technique. Third, the study of (Majewski et al., 2006) indicates that the injury risk is greater when the opponent adopts a certain grip style different from that of the norm. We did not consider grip style as a factor in our study.

Perspective

In the context of injury prevention research in judo, we recommend a progressive stepwise approach: 1. synthesise the epidemiology of injuries in a specific sport, 2. determine the movement characteristics that leads to an injury, 3. analyse high-risk movements and situations to determine the parameters responsible for the primary injury mechanism, 4. create an evasive movement or intervention to reduce the risk of injury in elite sports, and 5. epidemiologically evaluate its efficiency with respect to injury prevention. In the current prospective study, with reference to (Akoto et al., 2017) and (Koshida et al., 2010), we showed that the risk for an ACL injury is elevated in the *common osoto-gari defence reaction* regarding articular knee joint trajectories, torques and muscle excitation. The *modified osoto-gari defence reaction* reduces critical biomechanical risk factors and results in higher hamstring EMG activity immediately after initial contact, which could result in a decreased risk of experiencing an ACL injury. Therefore, we recommend the implementation of the *modified osoto-gari defence reaction* in daily judo practice and the evaluation of its injury-preventive effect in training and in competition. We believe that this prevention model is applicable to other sports and can contribute significantly to reducing the incidence of injury in elite athletes.

Acknowledgments

The authors thank the German and Swiss National Judo Teams as well as Richard Trautmann for participating in this study. We also thank Ralf-Henning Schäfer for illustrating the defence reaction techniques in judo.

Description of Author Contributions

CL, RR, ML, JR and MM were involved in the acquisition of data. CL and RR were involved in the interpretation and analysis of the data and editing of the manuscript. CL, RR, JP, ML, MM, JR drafted the original manuscript. CL and RR conceived and designed the original project. All authors read and approved the final manuscript.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported there is no funding associated with the work featured in this article.

References

- Akoto, R., Lambert, C., Balke, M., Bouillon, B., Frosch, K. H., & Hoher, J. (2017). Epidemiology of injuries in judo: A cross-sectional survey of severe injuries based on time loss and reduction in sporting level. *British Journal of Sports Medicine*, 52(17), 1109–1115. <https://doi.org/10.1136/bjsports-2016-096849>
- Baker, R. L. F., Reay, J., & Sangeux, M. (2017). The conventional gait model - Success and limitations. In B. Müller & S. Wolf, (Eds.), *Handbook of human motion* (pp. 1-19). Springer. [10.1007/978-3-319-30808-1_25-2](https://doi.org/10.1007/978-3-319-30808-1_25-2).
- Beard, D. J., Kyberd, P. J., O'Connor, J. J., Fergusson, C. M., & Dodd, C. A. (1994). Reflex hamstring contraction latency in anterior cruciate ligament deficiency. *Journal of Orthopaedic Research*, 12(2), 219–228. <https://doi.org/10.1002/jor.1100120211>

- Bencke, J., Aagaard, P., & Zebis, M. K. (2018). Muscle activation during ACL injury risk movements in young female athletes: A narrative review. *Frontiers in Physiology*, 9(15), 445. <https://doi.org/10.3389/fphys.2018.00445>
- Benjamini, Y. & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.
- Benjamini, Y., & Yekutieli, D. (2005). False discovery rate-adjusted multiple confidence intervals for selected parameters. *Journal of the American Statistical Association*, 100(469), 71–81. <https://doi.org/10.1198/016214504000001907>
- BR, W. H. Integrating motion capture data into knowledge based ingress motion simulation tool. *Proceedings, digital human modeling symposium: Ann Arbor*, 2013.
- Dyhre-Poulsen, P., & Krogsgaard, M. R. (2000). Muscular reflexes elicited by electrical stimulation of the anterior cruciate ligament in humans. *Journal of Applied Physiology*, 89(6), 2191–2195. <https://doi.org/10.1152/jappl.2000.89.6.2191>
- Engebretsen, L., Soligard, T., Steffen, K., Alonso, J. M., Aubry, M., Budgett, R., Dvorak, J., Jegathesan, M., Meeuwisse, W. H., Mountjoy, M., Palmer-Green, D., Vanhegan, I., & Renström, P. A. (2013). Sports injuries and illnesses during the London summer olympic games 2012. *British Journal of Sports Medicine*, 47(7), 407–414. <https://doi.org/10.1136/bjsports-2013-092380>
- Fleming, B. C., Renstrom, P. A., Beynnon, B. D., Engstrom, B., Peura, G. D., Badger, G. J., Johnson, R. J. (2001). The effect of weightbearing and external loading on anterior cruciate ligament strain. *Journal of Biomechanics*, 34(2), 163–170. [https://doi.org/10.1016/S0021-9290\(00\)00154-8](https://doi.org/10.1016/S0021-9290(00)00154-8)
- Frey, A., Lambert, C., Vesselle, B., Rousseau, R., Dor, F., Marquet, L. A., Toussaint, J. F., & Crema, M. D. (2019). Epidemiology of judo-related injuries in 21 seasons of competitions in France: A prospective study of relevant traumatic injuries. *Orthopaedic Journal of Sports Medicine*, 7(5), 2325967119847470. <https://doi.org/10.1177/2325967119847470>
- Fukuda, Y., Woo, S. L., Loh, J. C., Tsuda, E., Tang, P., McMahon, P. J., & Debski, R. E. (2003). A quantitative analysis of valgus torque on the ACL: A human cadaveric study. *Journal of Orthopaedic Research*, 21(6), 1107–1112. [https://doi.org/10.1016/S0736-0266\(03\)00084-6](https://doi.org/10.1016/S0736-0266(03)00084-6)
- Grapar Zargi, T., Drobic, M., Vauhnik, R., Koder, J., & Kacin, A. (2017). Factors predicting quadriceps femoris muscle atrophy during the first 12 weeks following anterior cruciate ligament reconstruction. *Knee*, 24(2), 319–328. <https://doi.org/10.1016/j.knee.2016.11.003>
- Hermens, H. J., Freriks, B., Disselhorst-Klug, C., & Rau, G. (2000). Development of recommendations for SEMG sensors and sensor placement procedures. *Journal of Electromyography and Kinesiology*, 10(5), 361–374. [https://doi.org/10.1016/S1050-6411\(00\)00027-4](https://doi.org/10.1016/S1050-6411(00)00027-4)
- Hirokawa, S., Solomonow, M., Luo, Z., Lu, Y., & D'Ambrosia, R. (1991). Muscular co-contraction and control of knee stability. *Journal of Electromyography and Kinesiology*, 1(3), 199–208. [https://doi.org/10.1016/1050-6411\(91\)90035-4](https://doi.org/10.1016/1050-6411(91)90035-4)
- Hood, S., Ishmael, M. K., Gunnell, A., Foreman, K. B., & Lenzi, T. (2020). A kinematic and kinetic dataset of 18 above-knee amputees walking at various speeds. *Scientific Data*, 7(1), 150. <https://doi.org/10.1038/s41597-020-0494-7>
- Imamura, R. T., Hreljac, A., Escamilla, R. F., & Edwards, W. B. (2006). A three-dimensional analysis of the center of mass for three different judo throwing techniques. *Journal of Sports Science & Medicine*, 5(CSSI), 122–131 <https://pubmed.ncbi.nlm.nih.gov/24357984/>.
- Ishii, T., Ae, M., Suzuki, Y., & Kobayashi, Y. (2018). Kinematic comparison of the seoi-nage judo technique between elite and college athletes. *Sports Biomechanics*, 17(2), 238–250. <https://doi.org/10.1080/14763141.2017.1284256>
- Jennings, A. G., & Seedhom, B. B. (1994). Proprioception in the knee and reflex hamstring contraction latency. *The Journal of Bone and Joint Surgery. British Volume*, 76(3), 491–494. <https://doi.org/10.1302/0301-620X.76B3.8175863>
- Kaminski, T., & Gentile, A. M. (1986). Joint control strategies and hand trajectories in multijoint pointing movements. *Journal of Motor Behavior*, 18(3), 261–278. <https://doi.org/10.1080/00222895.1986.10735381>
- Kim, K. S., Park, K. J., Lee, J., & Kang, B. Y. (2015). Injuries in national Olympic level judo athletes: An epidemiological study. *British Journal of Sports Medicine*, 49(17), 1144–1150. <https://doi.org/10.1136/bjsports-2014-094365>
- Koga, H., Nakamae, A., Shima, Y., Iwasa, J., Myklebust, G., Engebretsen, L., Bahr, R., Krosshaug, T. (2010). Mechanisms for non-contact anterior cruciate ligament injuries: Knee joint kinematics in 10 injury situations from female team handball and basketball. *The American Journal of Sports Medicine*, 38(11), 2218–2225 doi:10.1177/0363546510373570.
- Koshida, S., Deguchi, T., Miyashita, K., Iwai, K., & Urabe, Y. (2010). The common mechanisms of anterior cruciate ligament injuries in judo: A retrospective analysis. *British Journal of Sports Medicine*, 44(12), 856–861. <https://doi.org/10.1136/bjsm.2008.051425>
- Kujala, U. M., Taimela, S., Antti-Poika, I., Orava, S., Tuominen, R., & Myllynen, P. (1995). Acute injuries in soccer, ice hockey, volleyball, basketball, judo, and karate: Analysis of national registry data. *BMJ*, 311(7018), 1465–1468. <https://doi.org/10.1136/bmj.311.7018.1465>
- Li, G., Rudy, T. W., Sakane, M., Kanamori, A., Ma, C. B., & Woo, S. L. (1999). The importance of quadriceps and hamstring muscle loading on knee kinematics and in-situ forces in the ACL. *Journal of Biomechanics*, 32(4), 395–400. [https://doi.org/10.1016/S0021-9290\(98\)00181-X](https://doi.org/10.1016/S0021-9290(98)00181-X)
- Madhavan, S., & Shields, R. K. (2011). Neuromuscular responses in individuals with anterior cruciate ligament repair. *Clinical Neurophysiology*, 122(5), 997–1004. <https://doi.org/10.1016/j.clinph.2010.09.002>
- Majewski, M., Susanne, H., & Klaus, S. (2006). Epidemiology of athletic knee injuries: A 10-year study. *Knee*, 13(3), 184–188. <https://doi.org/10.1016/j.knee.2006.01.005>
- Markolf, K. L., Burchfield, D. M., Shapiro, M. M., Shepard, M. F., Finerman, G. A., & Slauterbeck, J. L. (1995). Combined knee loading states that generate high anterior cruciate ligament forces. *Journal of Orthopaedic Research*, 13(6), 930–935. <https://doi.org/10.1002/jor.1100130618>
- Murata, N. (2005). From Jutsu to Do: The birth Kodokan Judo.
- Navacchia, A., Bates, N. A., Schilaty, N. D., Krych, A. J., & Hewett, T. E. (2019). Knee abduction and internal rotation moments increase ACL force during landing through the posterior slope of the Tibia. *Journal of Orthopaedic Research*, 37(8), 1730–1742. <https://doi.org/10.1002/jor.24313>
- Paschos, N. K., Gartzonikas, D., Barkoula, N. M., Moraiti, C., Paipetis, A., Matikas, T. E., & Georgoulis, A. D. (2010). Cadaveric study of anterior cruciate ligament failure patterns under uniaxial tension along the ligament. *Arthroscopy: The Journal of Arthroscopic & Related Surgery*, 26(7), 957–967. <https://doi.org/10.1016/j.arthro.2009.12.013>
- Podraza, J. T., & White, S. C. (2010). Effect of knee flexion angle on ground reaction forces, knee moments and muscle co-contraction during an impact-like deceleration landing: Implications for the non-contact mechanism of ACL injury. *Knee*, 17(4), 291–295. <https://doi.org/10.1016/j.knee.2010.02.013>
- Pucsek, J. M., Nelson, K., & Ng, E. D. (2001). A kinetic and kinematic analysis of the Harai-goshi judo technique. *Acta Physiologica Hungarica*, 88(3–4), 271–280. <https://doi.org/10.1556/APhysiol.88.2001.3-4-9>
- Renstrom, P., Arms, S. W., Stanwyck, T. S., Johnson, R. J., & Pope, M. H. (1986). Strain within the anterior cruciate ligament during hamstring and quadriceps activity. *The American Journal of Sports Medicine*, 14(1), 83–87. <https://doi.org/10.1177/036354658601400114>
- Roelants, M., Verschueren, S. M., Delecluse, C., Levin, O., & Stijnen, V. (2006). Whole-body-vibration-induced increase in leg muscle activity during different squat exercises. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 20(1), 124–129. <https://doi.org/10.1519/R-16674.1>
- Ruas, C. V., Pinto, R. S., Haff, G. G., Lima, C. D., Pinto, M. D., & Brown, L. E. (2019). Alternative methods of determining hamstrings-to-quadriceps ratios: A comprehensive review. *Sports Medicine - Open*, 5(1), 11. <https://doi.org/10.1186/s40798-019-0185-0>
- Scattone-Silva GCL, R., Serrão, F. V., Serrão, F. V., & Serrão, F. V. (2012). Acceleration time, peak torque and time to peak torque in elite karate athletes. *Science & Sports*, 27(4), e31–e7. <https://doi.org/10.1016/j.scispo.2011.08.005>

- Serpell, B. G., Scarvell, J. M., Pickering, M. R., Ball, N. B., Newman, P., Perriman, D., Warmenhoven, J., & Smith, P. N. (2015). Medial and lateral hamstrings and quadriceps co-activation affects knee joint kinematics and ACL elongation: A pilot study. *BMC Musculoskeletal Disorders*, 16(1), 348. <https://doi.org/10.1186/s12891-015-0804-y>
- Shin, C. S., Chaudhari, A. M., & Andriacchi, T. P. (2011). Valgus plus internal rotation moments increase anterior cruciate ligament strain more than either alone. *Medicine & Science in Sports & Exercise*, 43(8), 1484–1491. <https://doi.org/10.1249/MSS.0b013e31820f8395>
- Simonsen, E. B., Magnusson, S. P., Bencke, J., Naesborg, H., Havkrog, M., Ebstrup, J. F., Sorensen, H. (2000). Can the hamstring muscles protect the anterior cruciate ligament during a side-cutting maneuver? *Scandinavian Journal of Medicine & Science in Sports*, 10(2), 78–84. <https://doi.org/10.1034/j.1600-0838.2000.010002078.x>
- Tsuda, E., Okamura, Y., Otsuka, H., Komatsu, T., & Tokuya, S. (2001). Direct evidence of the anterior cruciate ligament-hamstring reflex arc in humans. *The American Journal of Sports Medicine*, 29(1), 83–87. <https://doi.org/10.1177/03635465010290011801>
- Utturkar, G. M., Irribarra, L. A., Taylor, K. A., Spritzer, C. E., Taylor, D. C., Garrett, W. E., & DeFrate, L. E. (2013). The effects of a valgus collapse knee position on in vivo ACL elongation. *Annals of Biomedical Engineering*, 41(1), 123–130. <https://doi.org/10.1007/s10439-012-0629-x>
- Wiley, M. E., & Damiano, D. L. (1998). Lower-extremity strength profiles in spastic cerebral palsy. *Developmental Medicine & Child Neurology*, 40(2), 100–107. <https://doi.org/10.1111/j.1469-8749.1998.tb15369.x>